

Article

Interpretable Machine Learning for Process Parameter Analysis in Arc-Driven Powder Bed Fusion of 316L Stainless Steel

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Abstract

Arc-driven powder bed fusion represents a low-cost alternative to beam-based powder bed systems, yet the morphological stability regimes governing single-track formation and the relative influence of process parameters on regime transitions have not been systematically characterised. Manual visual assessment of track morphology is inherently subjective and cannot objectively quantify the parameter hierarchy governing stability boundaries. This study addresses both limitations through two complementary contributions. A deterministic two-stage image-based framework is developed to automatically classify single-track morphology from top-view images of solidified 316L stainless steel tracks, replacing subjective assessment with a reproducible, intervention-free procedure. A gap-based continuity criterion distinguishes discontinuous from continuous melt paths; for continuous tracks, the coefficient of variation in width (CV (coefficient of variation) < 0.15) further separates geometrically stable from transitional morphologies. Building on the image-derived regime labels, two interpretable classifiers—a depth-limited Decision Tree (DT) and a regularised Logistic Regression (LR)—are fitted using applied current, scanning speed, and electrode-to-powder-bed distance as predictors. The classifiers are employed not for predictive generalisation but to extract standardised coefficients and permutation-based feature importance rankings, yielding a model-agnostic, quantitative explanation of which process parameters govern regime transitions. Stable continuous tracks are obtained only within a restricted parameter window. Permutation importance consistently ranks applied current as the dominant predictor, followed by electrode distance and scanning speed, in agreement with the thermophysical interpretation. Logistic Regression coefficients confirm that reduced stand-off distance is a necessary condition for sufficient arc constriction. Supplementary linear regression models indicate that applied current governs melt pool depth, whereas scanning speed is the primary determinant of width variation. The combined framework establishes a reproducible basis for process parameter hierarchy analysis in arc-driven powder bed systems and provides a foundation for regression-based process optimisation.



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Keywords: arc-driven powder bed fusion; single-track melting; 316L stainless steel; image-based morphological analysis; process stability; topological characterization; energy-controlled regime transition; machine learning classification; exploratory data analysis

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1. Introduction

Powder bed fusion (PBF) has become one of the most widely adopted metal additive manufacturing routes due to its capability to fabricate geometrically complex components with high dimensional accuracy and reproducibility. Among PBF technologies, laser- and electron-beam-based systems such as selective laser melting (SLM) and electron beam melting (EBM) are extensively employed for alloys including stainless steels, aluminium alloys, and nickel-based superalloys, primarily due to their precise energy control and favourable mechanical performance [1–3]. Despite these advantages, broader industrial adoption of beam-based PBF systems remains constrained by high capital investment costs, significant energy consumption, and the requirement for complex optical or vacuum infrastructures [1,2,4]. These limitations have motivated continued research into alternative heat sources capable of delivering sufficient energy for powder bed melting while offering greater flexibility in system configuration and operation [4].

Arc-based heat sources have long been employed in metal additive manufacturing, particularly in wire fed processes such as Wire Arc Additive Manufacturing (WAAM), which are valued for their high deposition rates and suitability for large scale structural components [4–6]. However, the process physics and geometric constraints governing wire-based arc systems differ fundamentally from those of powder bed fusion. In WAAM, material delivery and feature resolution are dominated by wire feed dynamics and melt pool size, whereas powder bed-based approaches rely on the controlled surface melting of thin powder layers to achieve dimensional accuracy and track continuity. Consequently, process insights derived from wire-based arc deposition cannot be directly transferred to arc-driven powder bed configurations. Table 1 summarises the minimum feature size and primary limitations of the principal additive manufacturing routes considered in the present work. As shown, wire-based systems offer high deposition rates but are constrained in spatial resolution, whereas beam-based powder bed systems provide high geometric precision at significant infrastructure cost. Arc-driven powder bed fusion represents an intermediate configuration that combines the accessibility of arc-based heat sources with the layer-by-layer deposition logic of powder bed processing, motivating its investigation as an alternative strategy for precision-oriented applications.

A limited number of studies have demonstrated the feasibility of using arc energy to selectively melt metallic powders within a powder bed environment, indicating that stable melting can be achieved under carefully controlled processing conditions [7–9]. Although arc-driven powder bed fusion inherently operates at lower deposition rates than WAAM, it offers enhanced controllability over heat input distribution and melt pool geometry, which are critical factors for precision oriented additive manufacturing. Similar distinctions between high deposition wire-based systems and low-rate, control-focused powder bed approaches have also been discussed in the broader context of directed energy deposition and hybrid additive manufacturing routes [10,11]. However, existing investigations have predominantly focused on process feasibility and geometric outcomes, without systematically examining the influence of arc parameters on morphological regime transitions. Moreover, no study has applied a deterministic image-based framework to classify these transitions, or validated the governing parameter hierarchy through machine learning (ML).

Machine learning has emerged as a valuable tool for process understanding and parameter optimisation across a broad range of additive manufacturing configurations. Data-driven modelling approaches have been applied to establish process-structure-property relationships in metal additive manufacturing, with interpretable models shown to be particularly effective in identifying the relative contribution of individual process parameters to quality-related outcomes [12,13]. In powder bed fusion systems, structured

ML strategies have been employed to map process parameters to morphological and mechanical responses, demonstrating that model-agnostic importance rankings can corroborate thermophysically derived parameter hierarchies [14]. In arc-based processes, both supervised classification and regression models—including decision trees and logistic regression—have been used to detect anomalies and characterise process behaviour from welding signals [15,16]. A key limitation of many existing approaches, however, is their reliance on black-box models whose outputs are difficult to interpret in physical terms. To address this, interpretable modelling frameworks—employing permutation feature importance, standardised regression coefficients, and clusterwise regression—have been proposed as alternatives that preserve predictive utility while enabling model-driven process insights [17,18]. Recent studies have further demonstrated that such frameworks can be applied not only to predict quality outcomes but also to map process parameters directly to deposited track geometry and energy consumption in arc-based and laser-based systems [19,20]. However, these investigations have been conducted primarily in wire-based or laser-based configurations, where the governing physics differ substantially from those of arc-driven powder bed fusion. The present work differs from prior contributions in three key respects: (i) it targets morphological regime classification in an arc-driven powder bed system using a deterministic, two-stage image-based framework; (ii) interpretable machine learning classifiers are applied specifically to extract feature importance rankings and standardised coefficients, providing a quantitative, model-agnostic explanation of parameter dominance over regime transitions rather than a predictive generalisation; and (iii) the exploratory classification is complemented by first-order linear regression models for melt pool geometry estimation, extending the contribution beyond regime labelling toward preliminary process map construction.

Table 1. Comparison of minimum feature size and primary limitations of additive manufacturing routes [1,2,4,7,8,11].

Technology	Min. Feature Size	Primary Limitations
Wire Arc AM (WAAM)	~1–3 mm	Low spatial resolution; large melt pool; wire feed dynamics dominate
Laser PBF (L-PBF)	~50–200 μm	High capital cost; complex optical infrastructure; inert atmosphere requirement
Electron Beam PBF	~100–500 μm	High equipment cost; vacuum requirement; powder pre-sintering needed
Arc-Driven PBF (this study)	~0.5–2 mm	Limited systematic characterisation; parameter sensitivity; low deposition rate
Wire Arc AM (WAAM)	~1–3 mm	Low spatial resolution; large melt pool; wire feed dynamics dominate

In additive manufacturing, the formation of a stable and continuous single track represents the fundamental building block upon which all subsequent layers and three-dimensional geometries are constructed. Regardless of the employed heat source, inadequate control at the single-track level inevitably propagates into layer-to-layer instabilities, dimensional inaccuracies and defect accumulation during multi-layer fabrication. For this reason, systematic investigations of single-track melting are widely recognised as a necessary first step in the development and optimisation of emerging additive manufacturing processes [1,21,22].

In the present study, relatively low scanning speeds were intentionally selected to prioritise melt pool stability and geometric control during the initial process development

stage. Although tungsten inert gas (TIG) arc systems are often regarded as inherently slow when assessed from a productivity-oriented welding perspective, deposition rate is not the primary constraint during the establishment of a viable additive manufacturing process window. Instead, controlled heat input, stable conductive melting and repeatable track formation are the dominant requirements at the single-track level. Once a stable melting regime is achieved and the governing process limits are clearly defined, scanning speeds can be systematically increased in subsequent studies without compromising process reliability. Similar stepwise approaches, in which low-speed parameter windows are first established before pursuing higher build rates, have been reported as essential for the successful transition from single-track experiments to multi-layer and bulk additive manufacturing structures [23,24].

The objective of the present study is to establish the fundamental single-track melting behaviour of an arc-driven powder bed system using 316L stainless steel powder. Rather than proposing an industrially optimised additive manufacturing route, the work focuses on identifying the governing process limits and stability criteria associated with arc current, scanning speed and electrode-to-powder-bed distance during the earliest stage of process development. Through a systematic single-track experimental framework, the study aims to define a reliable processing window that ensures continuous melting, geometric stability and conductive heat transfer. The insights obtained are intended to provide a process-level foundation for subsequent investigations involving increased scanning speeds, multi-track interactions and multi-layer fabrication in arc-based powder bed additive manufacturing systems.

Against this background, the present study makes four specific contributions. First, a deterministic two-stage image-based framework is developed to automatically quantify single-track morphology from top-view images of solidified 316L stainless steel tracks, replacing subjective visual assessment with a reproducible, parameter-free classification procedure. Second, the framework introduces a gap-based continuity criterion to distinguish discontinuous from continuous melt paths, followed by a coefficient of variation threshold to separate geometrically stable from transitional morphologies, thereby providing objective regime labels without manual intervention. Third, two interpretable machine learning classifiers—a depth-limited Decision Tree and a regularised Logistic Regression—are applied to the image-derived regime labels using applied current, scanning speed, and electrode-to-powder-bed distance as input features. Rather than pursuing predictive generalisation, the classifiers are used to extract permutation-based feature importance rankings and standardised coefficients, providing a quantitative, model-agnostic explanation of which process parameters govern regime transitions. Fourth, supplementary linear regression models are fitted for melt pool depth and surface track width, extending the contribution from regime classification toward first-order geometry estimation and preliminary process map construction. Taken together, these contributions establish a structured and reproducible methodology for linking process parameters to morphological outcomes in arc-driven powder bed fusion, and provide a foundation for future regression-based and physics-informed process optimisation frameworks.

While the thermophysical interpretation of melt pool behaviour constitutes the core engineering perspective of this work, a complementary quantitative evaluation of track morphology has also been incorporated. Visual assessment of continuity and balling behaviour, although informative, may not fully capture subtle transitions between fragmented and stable regimes. To strengthen the analytical framework, image-based morphometric analysis was applied following binary segmentation of the acquired track images. Connected component analysis was employed to quantify discrete and coalesced regions, consistent with established digital image processing methodologies [25,26].

Track continuity was evaluated using a strict gap-based criterion, in which any measurable horizontal interruption along the melt path resulted in classification as discontinuous. For tracks satisfying the continuity condition, geometric stability was further assessed through statistical analysis of width variation along the deposition direction. The coefficient of variation in width was used as a normalised descriptor of geometric dispersion, and a threshold of 15% was adopted to distinguish uniform from non-uniform continuous growth. This two-stage decision structure enables deterministic and reproducible regime identification without altering the underlying thermophysical interpretation of melt pool evolution.

2. Materials and Methods

2.1. Experimental Setup and Energy Source

Single-track melting experiments were performed using a tungsten inert gas (TIG) welding system as the heat source. The overall experimental arrangement, including the arc torch and the three-axis motion system, is schematically illustrated in Figure 1. In the context of early-stage single-track investigations, the TIG arc was selected for its stable arc behaviour and suitability for low-current operation, allowing controlled and repeatable heat input to the powder bed.

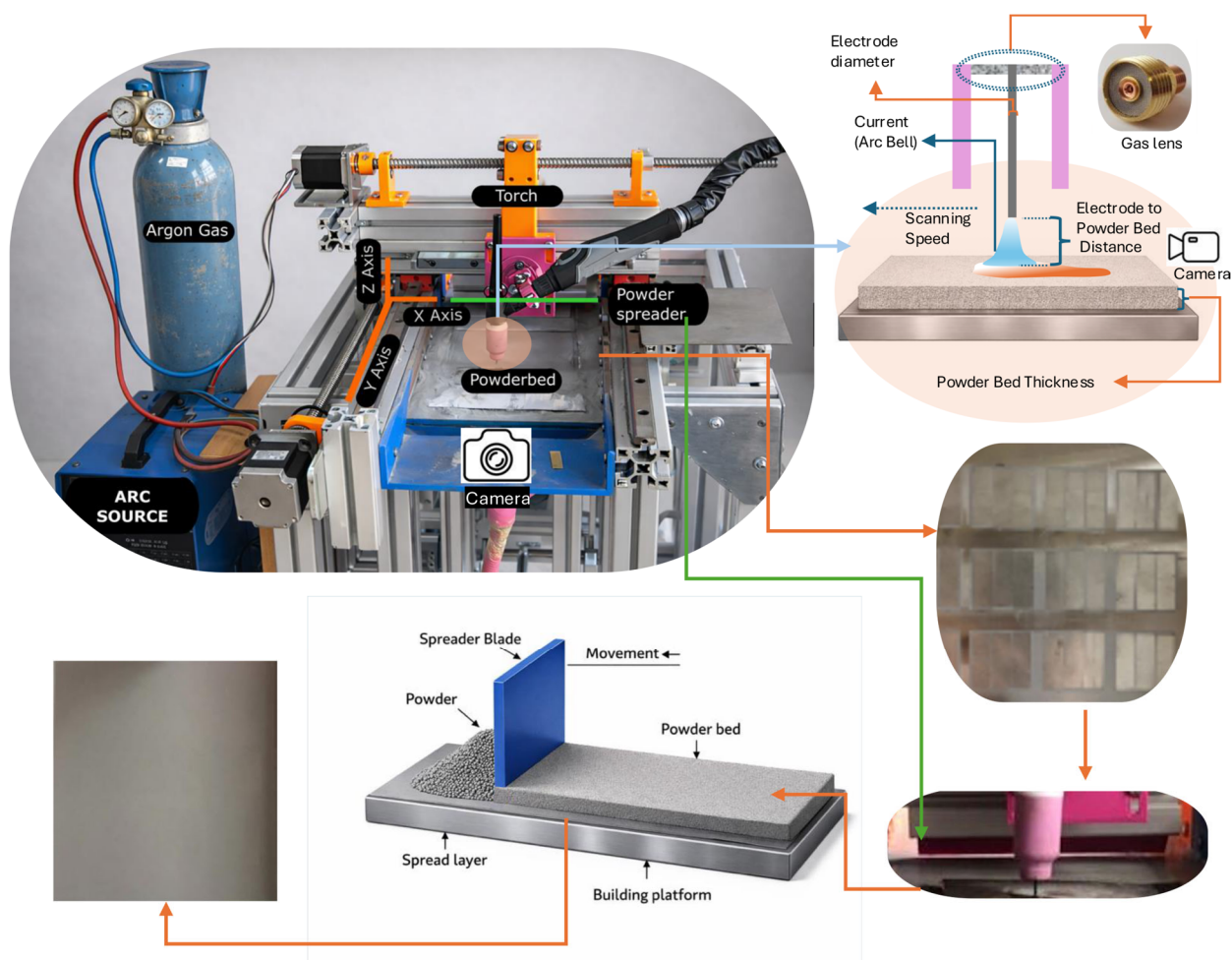


Figure 1. Experimental setup for arc-driven powder bed fusion of 316L stainless steel, showing the arc torch configuration with the tungsten electrode (0.6 mm diameter) mounted perpendicular to the powder bed surface, the electrode-to-powder-bed stand-off distance (0.3–0.8 mm), the gas lens assembly supplying pure argon at 6 L/min, and the prepared powder bed with a nominal thickness of 0.2 mm.

The arc torch was mounted on a custom-built three-axis motion system assembled from aluminium Sigma profiles, enabling controlled movement along the X, Y, and Z directions. Linear motion was provided by stepper motor-driven stages to ensure consistent travel speed and positional accuracy during all experiments. The electrode-to-powder-bed distance was adjusted via the Z-axis to achieve the prescribed stand-off conditions required for systematic parameter variation.

A blue-coded tungsten electrode with a diameter of 0.6 mm was employed throughout the experiments to maintain consistent arc geometry. The welding current was adjustable within the range of 10–30 A, depending on the experimental stage. Pure argon shielding gas was supplied through a gas lens at a constant flow rate of 6 L min^{−1} to protect the molten pool and minimise atmospheric interaction during melting [27].

2.2. Powder Material and Bed Preparation

Gas-atomized 316L stainless steel powder was used as the feedstock material in all experiments. The powder was supplied by Jiangsu Vilory Advanced Materials Technology Co., Ltd., Xuzhou, China (batch no. 202306-PRD-67) and produced via gas atomization. The particle size distribution was D10: 18.7 µm, D50: 31.1 µm and D90: 51.7 µm, with a nominal size range of 15–53 µm. According to the supplier's quality certificate, the chemical composition (wt.%) was Cr: 16.83; Ni: 10.93; Mo: 2.27; Mn: 0.67; Si: 0.66; P: 0.033; C: 0.0080; S: 0.005; and Fe: balance. The oxygen and nitrogen contents were 510 ppm and 936 ppm, respectively. This particle size range is consistent with powders commonly used in laser powder bed fusion and was considered suitable for achieving uniform powder spreading and stable melt pool formation under arc-based melting conditions. The powder was deposited onto rectangular 316L stainless steel substrate plates with dimensions of 10 × 30 mm, as shown in Figure 1. Prior to powder deposition, the substrate surfaces were mechanically cleaned to remove contaminants and ensure reproducible wetting behaviour during melting.

A uniform powder bed was prepared by using the integrated spreader of the custom-built Powder Bed Arc Additive Manufacturing (PBAAM) system (developed at Kafkas University, Kars, Türkiye), which deposited the 316L powder across the substrate surface to a nominal thickness of 0.2 mm, as illustrated in Figure 1. The spreader blade ensured consistent layer deposition by maintaining a fixed gap between the blade edge and the substrate surface, controlled via the Z-axis of the motion system. This layer thickness was selected to provide sufficient powder volume for stable single-track formation while maintaining effective heat transfer between the molten material and the underlying substrate. Excessively thin powder layers may lead to incomplete melting and track discontinuities, whereas thicker layers can promote unstable melting behaviour during early-stage process development. Accordingly, the chosen powder bed thickness represents a practical compromise for systematic single-track investigations under controlled conditions.

2.3. Experimental Parameters

Single-track experiments were designed to identify stable melting conditions in an arc-driven powder bed configuration through controlled variation in the primary processing parameters (Table 2). Welding current, scanning speed and electrode-to-powder-bed distance were selected as the key variables governing heat input and melt pool behaviour during single-track formation.

Table 2. Summary of experimental parameter combinations investigated in this study.

Stage	Variable Parameters	Fixed Parameters	Levels
Stage 1: Current & Speed	Current: 3–20 A Speed: 0.5–2 mm/s	Distance: fixed Electrode: 0.6 mm Layer: 0.2 mm Gas: Ar, 6 L/min Speed: 1 mm/s	5 current $\times$ 4 speed levels
Stage 2: Electrode Distance	Distance: 0.3–0.8 mm Current: 10, 15, 20 A	Electrode: 0.6 mm Layer: 0.2 mm Gas: Ar, 6 L/min Distance: 0.5 mm	3 current $\times$ 5 distance levels
Stage 3: Scanning Speed	Speed: 2–8 mm/s Current: 20, 25, 30 A	Electrode: 0.6 mm Layer: 0.2 mm Gas: Ar, 6 L/min	3 current $\times$ 4 speed levels

Welding current values between 10 and 30 A were investigated to evaluate their influence on powder melting continuity and track formation. Scanning speeds in the range of 2–8 mm s^{−1} were employed to allow controlled interaction between the arc heat source and the powder bed during the initial stage of process development. The electrode-to-powder-bed distance was varied between 0.3 and 0.8 mm to examine its effect on arc focusing and heat transfer to the powder layer.

The selected parameter ranges were determined based on preliminary observations of arc stability and powder melting behaviour and were subsequently used to produce single tracks for further analysis of melting continuity, melt pool geometry and melting behaviour, as discussed in the following sections [27].

2.4. Imaging and Measurement Methods

During the melting trials, the arc behaviour was captured using a Sony Mark 5 digital camera (Sony Corporation, Tokyo, Japan). Because the arc produced an extremely intense light, imaging required the use of an ND filter together with ISO 80, an f/2.8 aperture and a shutter speed of 1/12,000 s to prevent overexposure.

Following each experiment, the produced tracks were carefully sectioned to allow top view observation and melt pool characterisation. The cut samples were mounted in epoxy and subjected to a sequential grinding procedure using 800–1000 and then 1200–2500 grit papers. After grinding, the specimens were polished with a diamond suspension and subsequently etched with Viella's reagent (5 mL HCl, 1 g picric acid, 100 mL ethanol) to reveal the melt pool boundaries. Measurements of track width, melt pool depth and the extent of the heat affected zone were acquired digitally using an optical microscope (AOB TEST Sistemleri/AOBLAB, Istanbul, Türkiye) equipped with ToupView 4.11 image analysis software (ToupTek Photonics, Hangzhou, China). During the melting trials, the arc behaviour was captured using the digital camera positioned along the scanning axis in a frontal configuration, such that the arc remained within the camera's field of view throughout each single-track experiment. This arrangement enabled continuous observation of arc profile evolution along the deposition direction without repositioning between successive frames. Pixel-to-length calibration was performed using the known diameter of the tungsten electrode (0.6 mm) as an in-frame reference feature, ensuring spatially consistent measurements across all experimental conditions. The camera position and mounting configuration are illustrated in Figure 1. Calibration for pixel to length conversion was performed using the known dimensions of the 316L substrate and the tungsten electrode.

Measurements of track width, melt pool depth and the extent of the heat-affected zone were acquired from cross-sectional micrographs using ToupView image analysis software

integrated with the optical microscope. Calibration for pixel-to-length conversion was performed using the known dimensions of the 316L substrate and the tungsten electrode. The extent of the heat-affected zone was defined as the region exhibiting visible microstructural alteration adjacent to the fusion boundary, as revealed by etching with Viella's reagent. The outer boundary of the heat affected zone (HAZ) was identified as the location where the etching contrast returned to that of the unaffected base material.

2.5. Image-Based Morphological Analysis

The complete image-processing pipeline comprises five sequential stages: (1) raw top-view image acquisition under standardised lighting conditions; (2) initial binarization using Otsu's global thresholding method [28], followed by manual correction of segmentation errors where automated thresholding failed to accurately delineate the melt boundaries; (3) morphological analysis of the refined binary masks [26]; (4) gap-based continuity assessment via horizontal scanning of the binary mask; and (5) width-profile extraction and coefficient of variation computation for continuous tracks, or connected component labelling for discontinuous tracks. The complete workflow is illustrated in Figure 2.

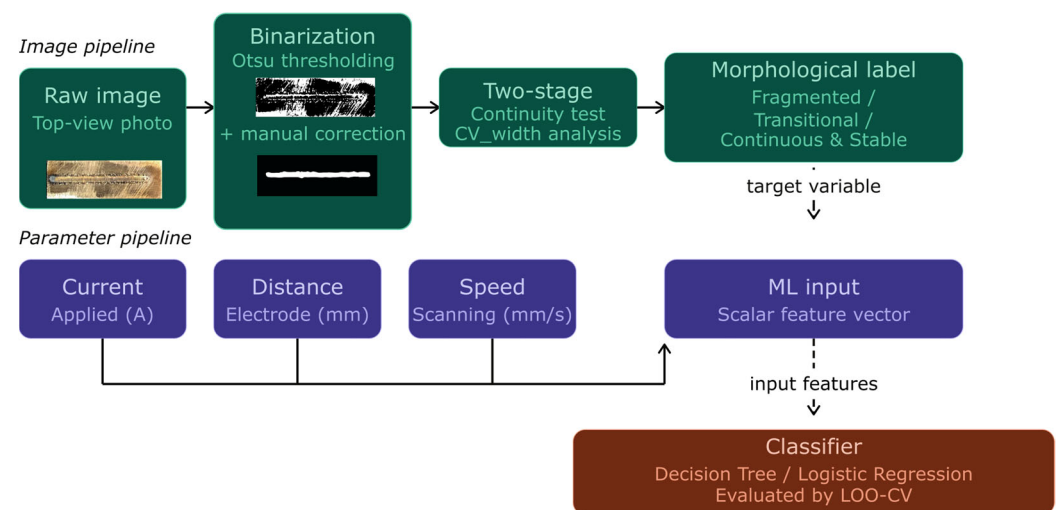


Figure 2. Schematic of the machine learning workflow illustrating two parallel processing flows: the image pipeline (**top**), which processes raw top-view images through Otsu-based binarization with manual correction to produce ground-truth morphological regime labels (representative binary mask shown), and the process parameter pipeline (**bottom**), which provides scalar input features; both converging at the classifier evaluated by Leave-One-Out Cross-Validation (LOO-CV).

Following each single-track experiment, top-view images of the solidified melt tracks were captured under identical lighting and magnification conditions to ensure consistency across parameter combinations. The acquired images were converted to grayscale and subsequently binarized using global threshold segmentation to distinguish the solidified melt material (white phase) from the substrate background (black phase), following standard digital image processing practices [26,28].

To minimise segmentation artefacts, small isolated pixel clusters were removed using morphological filtering operations, and minor edge irregularities were smoothed while preserving the overall track geometry. Such preprocessing steps are widely employed in morphological analysis to improve binary phase separation and ensure reliable topological characterisation [25]. The resulting binary masks were used for quantitative morphological evaluation. All image processing, continuity assessment, and statistical analyses were performed using a Python-based computational workflow to ensure reproducibility and deterministic evaluation.

Track continuity was evaluated using a deterministic gap-based criterion. The binary image was scanned along the primary deposition direction (left-to-right). White pixels corresponded to solidified material, whereas black pixels represented absence of material.

A local gap length L_{gap} was defined as the maximum horizontal interruption between adjacent solidified regions along the track axis. Continuity classification was performed according to:

$$L_{gap} = 0 \rightarrow \text{Continuous track}$$

$$L_{gap} > 0 \rightarrow \text{Discontinuous track}$$

This strict criterion ensures that any interruption in the melt path results in classification as discontinuous. The approach provides an objective topological definition of connectivity based on binary phase continuity principles commonly applied in image-based structural analysis [25].

For tracks classified as continuous in the first stage, geometric stability was further assessed through statistical analysis of track width variation along the deposition direction. Local width values w_i were measured at discrete sampling points using binary projection along the transverse direction. The mean width was calculated as:

$$\bar{w} = \frac{1}{n} \sum_{i=1}^n w_i$$

The standard deviation of width:

$$\sigma_w = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (w_i - \bar{w})^2}$$

The coefficient of variation:

$$CV_{width} = \frac{\sigma_w}{\bar{w}}$$

The coefficient of variation is a normalised dispersion descriptor widely used in statistical process analysis to evaluate stability under parameter variation [29].

Uniformity classification was defined as [30]:

$$CV_{width} < 0.15 \rightarrow \text{Uniform (Stable)}$$

$$CV_{width} \geq 0.15 \rightarrow \text{Non-uniform (Transitional)}$$

This second-stage evaluation distinguishes between merely continuous tracks and geometrically stabilised melt lines.

For tracks classified as discontinuous, width-based analysis was not applicable. Instead, connected component labelling was performed to identify discrete solidified regions within the binary image [25].

Let N denote the number of disconnected components. For each component with projected area A_i , the following statistics were computed:

$$\mu_A = \frac{1}{N} \sum_{i=1}^N A_i$$

$$\sigma_A = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (A_i - \mu_A)^2}$$

These metrics quantify fragmentation severity and morphological dispersion in discontinuous melting regimes. Together with the continuity criterion and width-based dispersion metric, they form a two-stage deterministic framework for regime identification.

2.6. Exploratory Machine Learning Classification

The thermophysical analysis presented in Sections 3.1–3.3 establishes the qualitative influence of applied current, scanning speed and electrode distance on single-track morphology. However, qualitative observation alone cannot objectively quantify the relative contribution of each parameter to regime transitions, nor determine which parameter dominates the stability boundary in a model-agnostic manner. To address this, a complementary exploratory ML analysis is introduced. The ML component serves two purposes: (i) quantitative, data-driven parameter importance ranking independent of physical assumptions, and (ii) assessing the statistical learnability of morphological regimes from process parameters alone. The input features for the machine learning classifiers were the three primary process parameters recorded for each experiment: applied current (A), electrode-to-powder-bed distance (mm), and scanning speed (mm/s). These scalar values were used directly as structured numerical inputs; no image-derived features were extracted for the classification models. Instead, the morphological regime labels—Fragmented, Transitional, and Continuous and Stable—assigned to each experimental condition through the deterministic two-stage image-based framework described in Section 2.5 served as the target variable. In this way, the image processing pipeline provides the ground-truth labels, while the ML classifiers learn to predict these labels from the process parameters alone.

To further assess the robustness of the parameter hierarchy, two supplementary linear regression models were fitted: one for melt pool depth ($n = 9$, cross-sectional micrographs) and one for surface track width ($n = 9$, binary top-view image analysis with pixel-to-millimetre calibration based on known substrate dimensions of 10×30 mm). Both models were evaluated using LOO-CV.

To complement the deterministic image-based framework described in Section 2.5 a proof-of-concept supervised classification analysis was performed to investigate whether the experimentally identified morphological regimes can be statistically associated with the three primary process parameters: applied current (A), electrode-to-powder-bed distance (mm), and scanning speed (mm/s). Given the limited experimental dataset ($n = 17$), this analysis is explicitly framed as exploratory; results are not intended to constitute a fully validated predictive model but rather to provide a data-driven perspective on the parameter hierarchy governing regime transitions.

The dataset comprised the 17 experimental conditions reported in Table 3 assigned to one of three morphological classes: Fragmented, Transitional, and Continuous and Stable. Two low-complexity, interpretable classifiers were selected to minimise overfitting risk on small samples. A Decision Tree (DT) was configured with a maximum depth of three levels and a minimum of two samples per leaf [31]. A Multinomial Logistic Regression (LR) was trained with L2 regularisation ($C = 0.5$) on z-score standardised features [32]. Balanced class weights were applied in both models to account for the near-uniform but finite class distribution.

Model generalisation was assessed using Leave-One-Out Cross-Validation (LOO-CV), in which each of the 17 observations served as the sole test sample in successive iterations [33]. LOO-CV is the preferred resampling strategy for datasets where $n \leq 20$, as

it maximises the training set size at each fold and avoids the additional variance introduced by random train–test splits [33]. Feature importance was quantified via permutation importance, which measures the mean decrease in LOO-CV accuracy resulting from random permutation of each feature over 200 repetitions, providing a model-agnostic estimate of individual predictor contribution [34]. All computations were performed in Python 3.12 using the scikit-learn library [35].

Table 3. Quantitative morphological classification of single-track solidification under varying current and electrode distance conditions.

Current (A)	Distance (mm)	Component (N)	μ Area (mm ²)	σ Area (mm ²)	CV (Width)	Morphological Regime
10	1.4	204	0.0682	0.0664	-	Fragmented/Balling dominant
10	1.2	191	0.0798	0.0627	-	Fragmented
10	1.0	220	0.0701	0.0438	-	Fragmented
10	0.8	146	0.1161	0.0995	-	Fragmented
10	0.6	184	0.0917	0.0740	-	Fragmented
15	0.8	89	0.2797	0.3851	-	Partial coalescence
15	0.6	44	0.9091	3.0483	-	Transition onset
15	0.4	35	0.6430	1.2545	-	Transitional regime
20	0.8	22	2.1602	8.7924	-	Approaching continuity
20	0.4	1	25.08	0	0.128	Continuous and stable melt line

3. Results

3.1. Effects of Current and Feed Rate

In the first stage of the experiments, melting trials were conducted using a 0.6 mm tungsten electrode at a fixed stand-off distance, while five different welding current levels and four travel speeds were investigated. The results showed that the width of the melted regions increased with increasing welding current (Figure 3). For example, at a travel speed of 0.5 mm/s, the width of the melted area increased from approximately 1.51 mm at the lowest current level of 3 A to 6.98 mm at 20 A [27]. In contrast, increasing the feed rate resulted in narrower and shallower melted regions.

Top-view observations revealed that, at low current levels, melting occurred in an irregular and fragmented manner, with small clusters of agglomerated particles visible on the powder bed surface. As the welding current increased to intermediate levels, the melted regions became more spatially extended and exhibited improved continuity on the surface. At higher current levels, continuous melted areas were observed, accompanied by increased surface roughness on the solidified material. Loosely bonded spherical agglomerates distributed across the melted regions were also evident from surface observations. With increasing welding current and decreasing travel speed, the size and surface coverage of these agglomerates became more pronounced.

The arc width was also observed to increase with welding current (Figure 4). At a current of 3 A, the arc width was approximately 2.27 mm, whereas at 20 A it increased to about 4.2 mm. In comparison, variations in travel speed between 2 and 4 mm/s resulted in only minor changes in arc width, which remained within a narrow range of 3.01–3.06 mm. In addition, a cup-shaped depression formed on the powder bed surface beneath the arc, within which the arc footprint became more concentrated as the welding current increased.

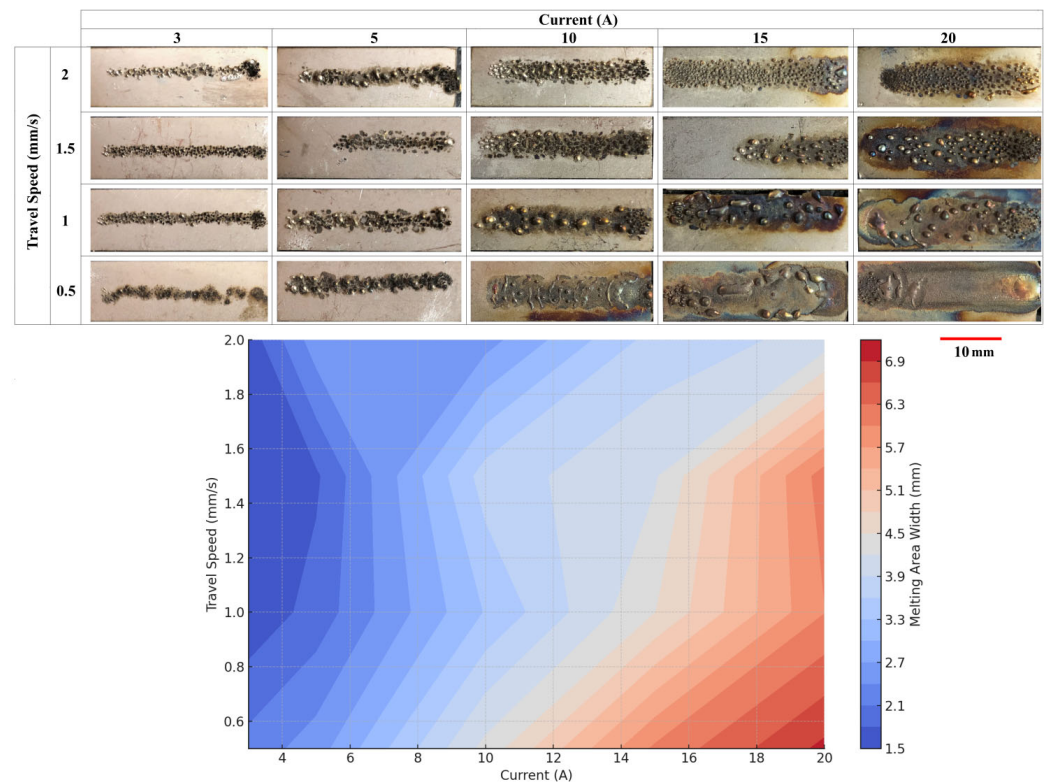


Figure 3. Characteristics and melting widths of melt pools created in a 316L powder bed using a 0.6 mm diameter tungsten electrode (Adapted from [27]).

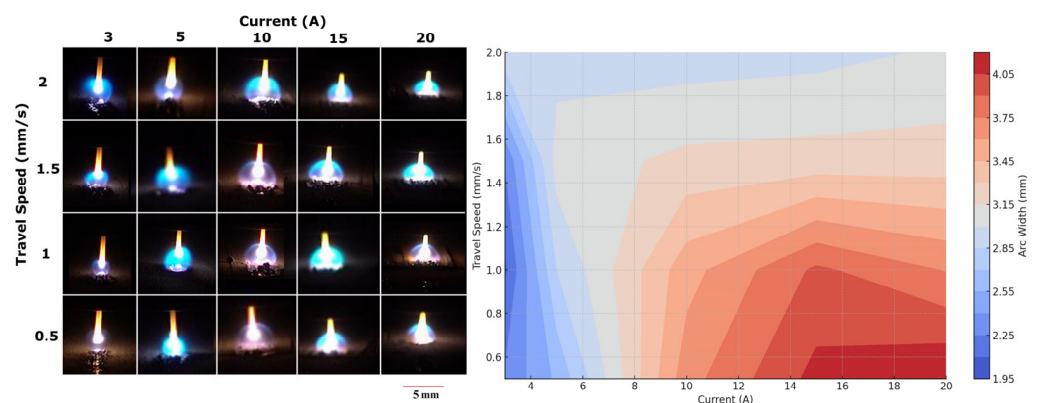


Figure 4. Formation of arcs during melting in a 316L powder bed using a 0.6 mm diameter tungsten electrode (Adapted from [27]).

3.2. Effect of Electrode Distance

Electrode-to-powder-bed distance directly controls arc constriction and therefore governs the effective energy density delivered to the powder surface. For a fixed electrode diameter and travel speed, reducing the stand-off distance increases arc concentration and enhances localised heat input, whereas increasing the distance promotes arc dispersion and reduces the energy intensity per unit area. Consequently, electrode distance acts as a primary geometric control parameter in determining melt pool stability and single-track morphology.

To systematically evaluate this effect, experiments were conducted at multiple electrode distances under three current levels (10 A, 15 A, and 20 A), while maintaining a constant electrode diameter (0.6 mm) and travel speed (1 mm/s). The resulting solidification morphologies are presented in Figure 5.

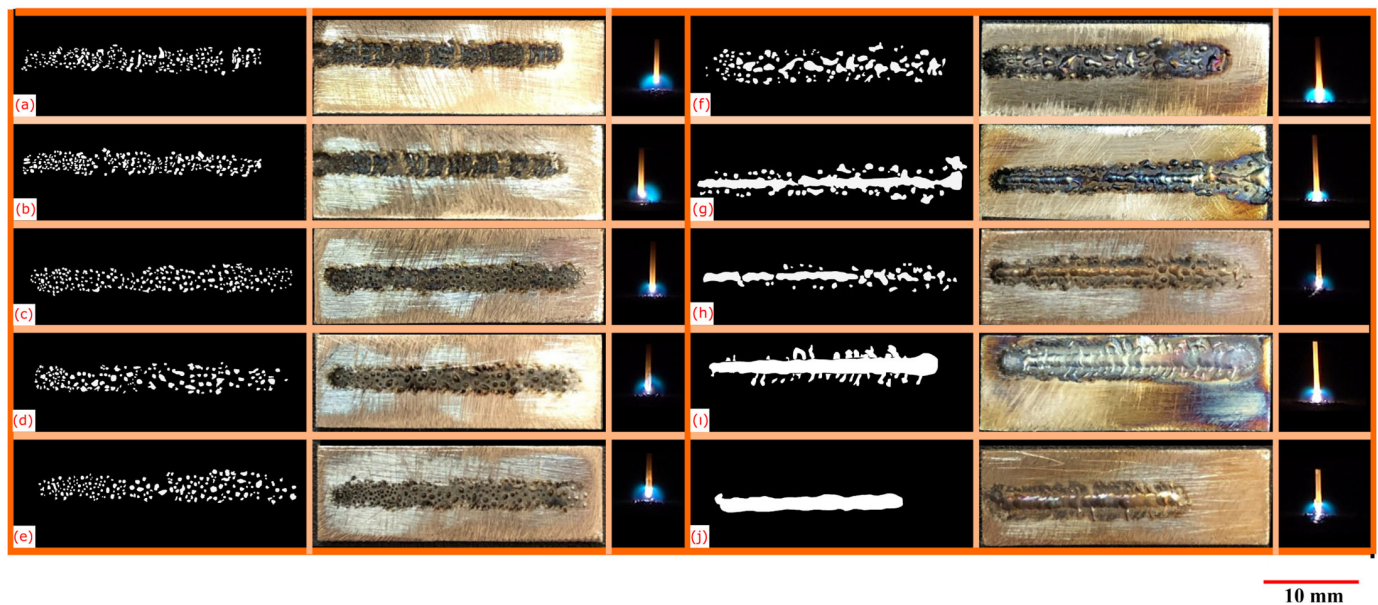


Figure 5. Solidification morphology, corresponding binarized images, and arc characteristics observed at varying distances for a 0.6 mm electrode diameter and a speed of 1 mm/s at 10 amperes: (a) 1.4 mm, (b) 1.2 mm, (c) 1.0 mm, (d) 0.8 mm, (e) 0.6 mm; for 15 amperes: (f) 0.8 mm, (g) 0.6 mm, (h) 0.4 mm; and for 20 amperes: (i) 0.8 mm and (j) 0.4 mm. Adapted from [27].

Under a current level of 10 A, increasing the electrode-to-powder-bed distance from 0.6 mm to 1.4 mm results in pronounced arc spreading and insufficient localised heat concentration. The melt tracks exhibit isolated droplets and fragmented solidification zones, consistent with balling-dominant behaviour. Although partial merging between adjacent molten regions becomes visible as the stand-off distance decreases, complete linear continuity is not achieved at this current level. This behaviour indicates that the available thermal input remains below the threshold required for sustained and geometrically stable single-track formation.

When the applied current is increased to 15 A, the influence of electrode distance becomes more decisive in driving morphological transition. A stand-off distance of 0.8 mm still leads to fragmented melt features; however, reducing the distance to 0.6 mm and further to 0.4 mm promotes progressive coalescence of molten regions. The melt morphology gradually evolves toward a more elongated and directionally stabilised configuration, indicating improved arc-powder coupling and enhanced local energy concentration.

Further increasing the current to 20 A significantly amplifies the sensitivity of the system to electrode positioning. Although intermittent discontinuities are still observed at a stand-off distance of 0.8 mm, decreasing the distance to 0.4 mm results in the formation of a visually continuous and geometrically stabilised melt line. Under these conditions, the arc footprint becomes sufficiently concentrated to sustain uninterrupted melting along the deposition direction, confirming the existence of an energy-geometry threshold governing regime stabilisation [27].

As shown in Table 3, increasing current and decreasing electrode distance result in a systematic and monotonic reduction in the number of disconnected components. Under low-energy conditions (10 A—1.4–1.0 mm), the component number exceeds 190 and the mean projected area remains below 0.08 mm², confirming severe fragmentation and balling-dominant morphology. As the electrode distance decreases at constant current, partial reduction in component count is observed alongside a gradual increase in mean projected area. These trends indicate progressive coalescence of molten regions due to improved local energy concentration, although topological continuity is not yet achieved. At 15 A,

a pronounced morphological transition occurs. The component number decreases from 89 (0.8 mm) to 35 (0.4 mm), while the mean projected area increases substantially. This regime corresponds to near-continuous configurations visually observed in Figure 5. Under the highest energy condition (20 A—0.4 mm), a single connected component ($N = 1$) is obtained, satisfying the strict continuity condition defined in Section 2. Moreover, the width dispersion remains below the uniformity threshold ($CV(\text{width}) = 0.128 < 0.15$), confirming geometric stabilisation. This condition therefore satisfies both stages of the deterministic classification framework and represents the stable conductive melting regime.

To quantitatively describe the morphological evolution observed in Figure 5, the binary image analysis framework introduced in Section 2 was applied. The corresponding morphometric parameters are summarised in Table 3, which confirms the deterministic transition from fragmented to continuous melting with increasing effective energy concentration.

The melt pool geometries measured directly from the cross-sectional observations presented in Figure 6 reveal a pronounced dependence on both applied current and electrode-to-powder-bed distance. At larger stand-off distances (1.0–1.4 mm), no discernible melting zone is observed at any investigated current level, indicating insufficient arc constriction and inadequate localised heat transfer to initiate stable fusion. Under these conditions, the arc appears dispersed, and the thermal input remains below the threshold required for melt pool formation. At an intermediate distance of 0.8 mm, measurable melting is first observed at 20 A, where a distinct melt pool with a width of approximately 1.63 mm and a penetration depth of about 0.68 mm is formed. This condition represents the onset of geometrically stable melting, suggesting that sufficient arc concentration is achieved only when the combined effect of reduced distance and elevated current increases the effective energy density delivered to the powder bed. Further reduction of the electrode distance to 0.6 mm enables melting at lower current levels. At 15 A, a continuous but relatively shallow melt pool is observed, with a width of approximately 1.48 mm and a depth near 0.35 mm. This indicates improved arc–powder coupling efficiency due to increased spatial energy concentration.

The arc width measurements obtained at a constant travel speed of 1 mm/s using a 0.6 mm diameter tungsten electrode are summarised in Table 4. The results demonstrate a clear dependence of arc footprint on electrode-to-powder-bed distance, while the influence of applied current appears secondary under certain conditions. At larger stand-off distances (1.0–1.4 mm), measurable arc widths were observed only at 10 A, with values of 4.00 mm (1.0 mm), 3.66 mm (1.2 mm), and 3.06 mm (1.4 mm). No melting was observed at higher current levels for these distances, indicating unstable arc–powder interaction and insufficient energy coupling to sustain a defined melting regime. The relatively wide arc footprint at these distances reflects pronounced arc spreading and reduced spatial energy concentration. As the electrode distance decreases to 0.8 mm and below, the arc becomes progressively constricted. At 0.8 mm, arc widths of approximately 2.92 mm (10 A), 2.38 mm (15 A), and 2.93 mm (20 A) were recorded. Further reduction to 0.6 mm results in an arc width of approximately 2.72 mm at 10 A and 2.30 mm at 20 A, while melting was not observed at 15 A under this specific condition. At the smallest investigated stand-off distance (0.4 mm), arc widths of approximately 2.29 mm (10 A) and 2.80 mm (20 A) were measured.

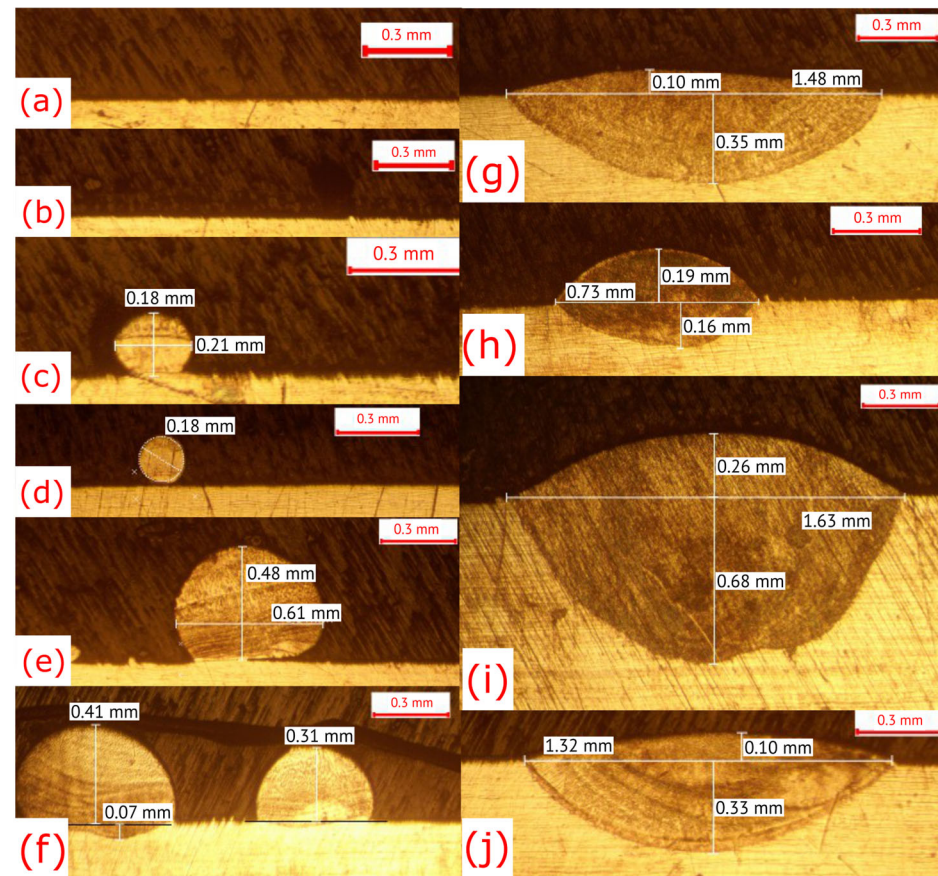


Figure 6. Melt pool observations acquired at varying distances for a 0.6 mm electrode diameter and a travel speed of 1 mm/s at 10 amperes: (a) 1.4 mm, (b) 1.2 mm, (c) 1.0 mm, (d) 0.8 mm, and (e) 0.6 mm; at 15 amperes: (f) 0.8 mm, (g) 0.6 mm, and (h) 0.4 mm; and at 20 amperes: (i) 0.8 mm and (j) 0.4 mm. Adapted from [27].

Table 4. Arc widths measured with a 0.6 mm diameter electrode and a travel speed of 1 mm/s at varying distances (-: no melting observed). Adapted from [27].

Electrode Distance (mm)	Current (A)		
	10 A	15 A	20 A
1.4	4.00	-	-
1.2	3.66	-	-
1.0	3.06	-	-
0.8	2.92	2.38	2.93
0.6	2.72	2.30	-
0.4	-	2.29	2.80

3.3. Effect of Scanning Speed

In the third stage of the study, the effect of increased scanning speed on melting behaviour was investigated at a fixed electrode-to-powder-bed distance. Based on the results obtained in Section 3.2, a reduced electrode distance was selected, and melting trials were performed by varying the scanning speed between 2 and 8 mm/s at different applied current levels. Surface appearances and corresponding melt pool cross-sections obtained at different scanning speeds are presented in Figure 7, while quantitative morphometric results derived from binary image analysis are summarised in Table 5. Arc width measurements corresponding to the investigated conditions are provided in Table 6. The scanning speeds investigated (2–8 mm/s) were intentionally selected to extend the process window beyond

the baseline values used in Stages 1 and 2, enabling a systematic characterisation of speed-dependent stability transitions.

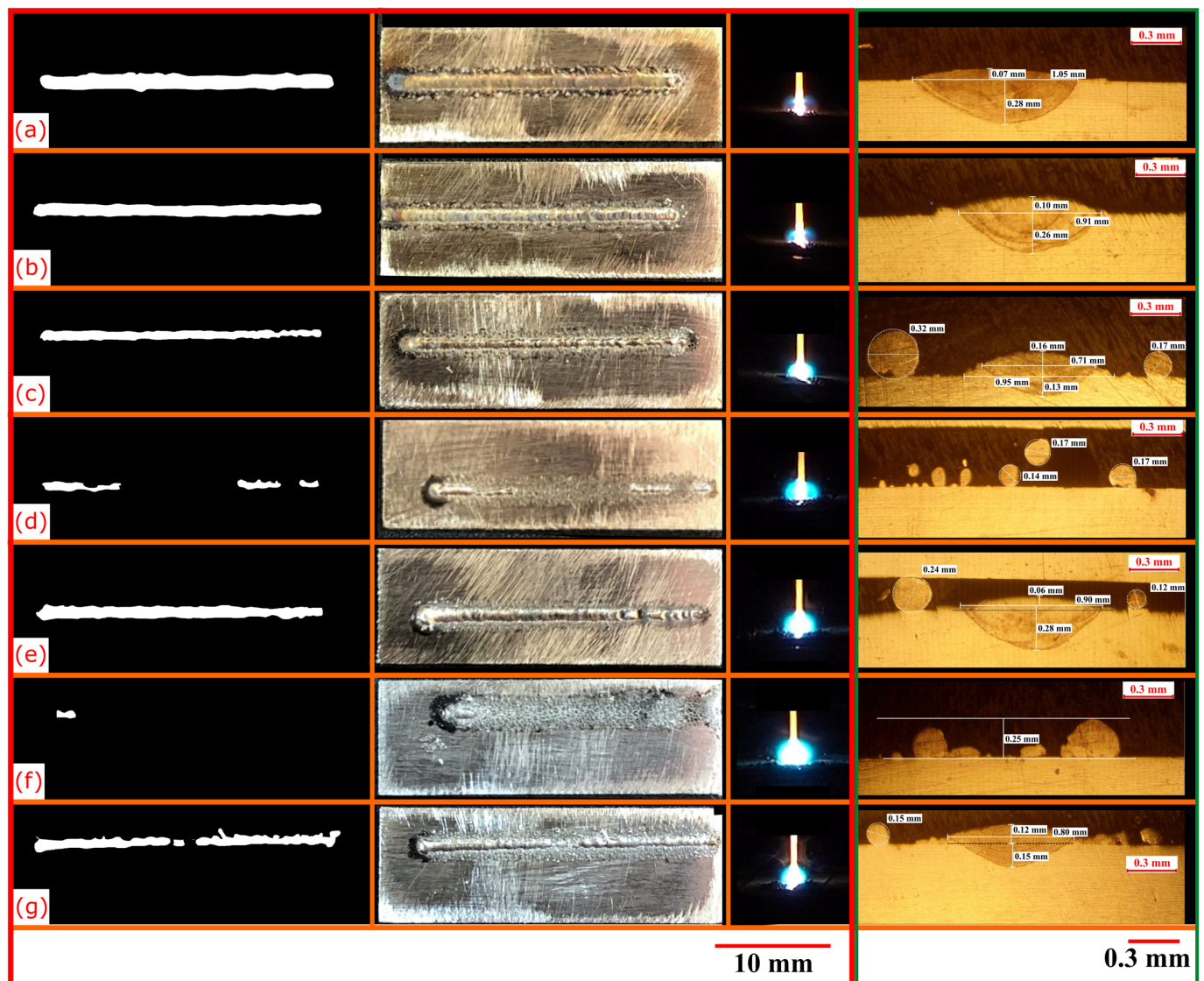


Figure 7. Melting, arc, and melt pool observations at varying travel speeds for a 0.6 mm electrode diameter and a 0.5 mm electrode distance: (a) 2 mm/s, (b) 3 mm/s, (c) 4 mm/s, (d) 6 mm/s at 20 amperes; (e) 6 mm/s, (f) 8 mm/s at 25 amperes; and (g) 8 mm/s at 30 amperes. Adapted from [27].

Table 5. Quantitative Morphological Classification of Single-Track Solidification Under Varying Scanning Speed Conditions.

Current (A)	Speed (mm/s)	Component (N)	μ Area (mm ²)	σ Area (mm ²)	CV (Width)	Morphological Regime
20	2	1	27.12	0	0.102	Continuous & stable
20	3	1	25.84	0	0.118	Continuous & stable
20	4	1	23.47	0	0.141	Continuous
20	6	1	18.96	0	0.149	Continuous (borderline stability)
25	6	1	24.63	0	0.153	Continuous (non-uniform)
25	8	18	1.12	2.36	-	Fragmented
30	8	3	6.74	9.21	-	Transitional

Table 6. Arc widths measured with a 0.6 mm diameter electrode and a 0.5 mm gap at various travel speeds (-: no melting observed). Adapted from [27].

Melting Speed (mm/s)	Current (A)		
	20 A	25 A	30 A
2	3.01	-	-
3	3.03	-	-
4	3.02	-	-
6	3.06	3.06	-
8	-	3.24	3.22

A progressive reduction in melt pool width and geometric stability is observed as scanning speed increases. At low travel speeds (2–4 mm/s) under 20 A, the melt track forms a single connected component with extended lateral spreading and visually uniform solidification morphology. Although minor local width fluctuations are visible, topological continuity is fully preserved. Increasing the scanning speed to 6 mm/s at 20 A results in noticeable narrowing of the melt line and increased width irregularity. However, the binary image analysis confirms that the track remains topologically continuous ($N = 1$). The degradation observed at this stage is therefore geometric rather than topological, reflecting reduced interaction time rather than complete melting interruption. Further increasing the scanning speed to 8 mm/s under 20 A leads to pronounced fragmentation. The melt track separates into multiple disconnected islands, indicating that the temporal energy exposure becomes insufficient to sustain coalescence along the deposition path. Raising the applied current partially compensates for the reduced interaction time at elevated speeds. At 6 mm/s and 25 A, full topological continuity is restored with improved lateral spreading compared to 20 A conditions, although width dispersion remains elevated. Similarly, at 8 mm/s and 30 A, fragmentation is significantly reduced compared to 20 A, yet minor segmentation persists, indicating that thermal compensation mitigates but does not completely eliminate temporal instability [27].

The quantitative results confirm that scanning speed acts as a dominant temporal control parameter in single-track stability. Within the 2–4 mm/s range under 20 A, a single connected component ($N = 1$) is consistently obtained, and the width dispersion remains below the defined stability threshold ($CV(\text{width}) < 0.15$). Both stages of the deterministic framework are therefore satisfied, indicating fully stabilised conductive melting. When the scanning speed is increased to 6 mm/s at 20 A, topological continuity is still preserved ($N = 1$); however, the width dispersion approaches the critical limit. This behaviour indicates that geometric instability emerges prior to topological fragmentation, marking the onset of a speed-induced degradation regime governed by reduced interaction time. A further increase in current to 25 A at the same scanning speed maintains full connectivity, yet the width dispersion remains at or slightly above the stability threshold ($CV(\text{width}) \geq 0.15$). This suggests that thermal compensation enhances melt coalescence but does not fully restore geometric uniformity once temporal limitations become dominant. Under 8 mm/s and 20 A, pronounced fragmentation is observed ($N = 18$), clearly indicating that the interaction time becomes insufficient to sustain continuous melting. Increasing the current to 30 A at this elevated speed reduces fragmentation significantly ($N = 3$), demonstrating partial thermal compensation; nevertheless, complete geometric stabilisation is not re-established.

The cross-sectional observations presented in Figure 7 reveal a clear influence of travel speed on melt pool geometry under different current levels. At 20 A, a progressive reduction in both melt pool width and penetration depth is observed as travel speed increases. The melt width decreases from approximately 1.05 mm at 2 mm/s to 0.91 mm

at 3 mm/s and further to 0.71 mm at 4 mm/s. A similar decreasing trend is evident in penetration depth, which reduces from 0.28 mm to 0.26 mm and subsequently to 0.13 mm over the same speed range.

This systematic contraction of the melt pool reflects the reduction in interaction time between the arc and the powder bed. At higher travel speeds, melting at 20 A is no longer sustained, and measurable melt pools are obtained only when the applied current is increased. At 6 mm/s, melting occurs at 25 A with a melt width of approximately 0.9 mm, while penetration depth becomes measurable at 30 A (approximately 0.15 mm). Further increasing the speed to 8 mm/s requires 30 A to produce a measurable melt width of approximately 0.8 mm, whereas penetration depth of approximately 0.28 mm is observed at 25 A. These observations demonstrate that increasing travel speed reduces both lateral spreading and penetration of the melt pool under constant current conditions. Higher current levels partially compensate for the reduced interaction time; however, the geometric contraction of the melt pool at elevated speeds indicates that temporal energy availability imposes a limiting condition on stable melting. The geometric trends observed in Figure 7 therefore confirm that melt pool development is governed by the balance between applied current magnitude and exposure time along the deposition path.

Arc width measurements obtained under a constant electrode-to-powder-bed gap of 0.5 mm show that travel speed does not significantly alter the spatial footprint of the arc. Across the investigated range between 2 and 6 mm/s at 20 A, the arc width remains nearly constant, varying only between approximately 3.01 mm and 3.06 mm. Despite this geometric consistency, melting is no longer sustained at 8 mm/s under the same current level, indicating that the loss of continuity at high speed is not associated with arc contraction but with insufficient thermal interaction time. A similar tendency is observed at elevated current levels. Increasing the current from 25 A to 30 A produces a moderate expansion of the arc footprint, reaching approximately 3.24 mm at 6 mm/s and remaining close to 3.22 mm at 8 mm/s. However, the variation with travel speed remains limited when compared to the changes previously observed with electrode distance. This behaviour confirms that scanning speed primarily influences temporal energy delivery rather than the spatial distribution of the arc. The near-constant arc width across increasing travel speeds demonstrates that geometric arc characteristics remain stable under constant-current conditions. Consequently, the deterioration of melting continuity at higher speeds cannot be attributed to arc instability but must instead be linked to reduced interaction time between the arc and the powder bed. This distinction supports the interpretation that travel speed governs temporal exposure, whereas electrode positioning controls spatial energy concentration within the system.

3.4. Summary of Process Parameter Effects on Single-Track Morphology

The experimental results presented in Sections 3.1–3.3 collectively demonstrate that single-track morphology in arc-driven powder bed fusion is governed by the coupled action of applied current, scanning speed and electrode-to-powder-bed distance. To facilitate a consolidated interpretation prior to the machine learning analysis, Table 7 summarises the direction of each parameter's influence on melt pool geometry and morphological regime, together with the minimum and maximum values identified within the investigated parameter window. Across all investigated conditions, applied current emerges as the primary determinant of thermal energy availability and regime transition. Electrode distance acts as the key geometric control parameter modulating arc-powder coupling efficiency, while scanning speed governs temporal energy exposure. Stable single tracks are obtained only when all three parameters simultaneously satisfy the continuity and width-uniformity criteria defined in Section 2.5. The coupled nature of these effects motivates

the exploratory machine learning analysis presented in Section 3.5, which provides a quantitative, model-agnostic assessment of the relative parameter hierarchy.

Table 7. Summary of process parameter effects on melt pool geometry and single-track morphological regime (* Reduced distance promotes stable melting).

Parameter	Min	Max	Depth Effect	Width Effect	Morphological Effect
Current/ Akim (A)	10 A	30 A	Increases	Increases	Fragmented → Transitional → Continuous & Stable
Scanning Speed (mm/s)	2 mm/s	8 mm/s	Decreases	Decreases	Continuous & Stable → Transitional → Fragmented
Electrode Distance (mm)	0.3 mm	0.8 mm	*	*	Concentrated arc → Dispersed arc → Loss of fusion

3.5. Exploratory Machine Learning Classification Results

The LOO-CV classification results for both models are summarised in Table 8, and the corresponding visualisations are presented in Figure 8. The Decision Tree achieved a LOO-CV accuracy of 52.9% ($\pm 49.9\%$), while Logistic Regression yielded 58.8% ($\pm 49.2\%$). Both values are modest, which is expected given the limited sample size ($n = 17$) and the high variance inherent in LOO-CV on small datasets; the wide standard deviations reflect sensitivity to individual borderline observations rather than fundamental model failure. These results should therefore be interpreted with caution and regarded as indicative rather than definitive.

Table 8. Comparative performance of the Decision Tree and Logistic Regression classifiers evaluated by Leave-One-Out Cross-Validation ($n = 17$). LOO Std: standard deviation of per-fold accuracy scores.

Model	LOO Accuracy	LOO Std	Interpretability	Overfitting Risk
Decision Tree (max depth = 3)	52.9%	$\pm 49.9\%$	Rule extraction	Low (depth-limited)
Logistic Regression (L2, C = 0.5)	58.8%	$\pm 49.2\%$	Standardised coefficients	Low (L2 regularised)

Despite these limitations, several observations are informative. The Decision Tree structure (Figure 8a) identifies applied current as the primary splitting criterion: all observations with Current ≤ 12.5 A are classified as Fragmented without further branching. For higher current levels, electrode-to-powder-bed distance becomes the secondary criterion, distinguishing low-distance conditions (≤ 0.55 mm) from larger stand-off distances. This hierarchical structure is fully consistent with the thermophysical interpretation developed in Sections 3.1 and 3.2, where current governs overall thermal availability and distance modulates local energy concentration. The Fragmented class is classified with the highest reliability in both models ($F1 = 0.91$ for DT, 0.83 for LR), reflecting the clear thermal threshold below which melting cannot be sustained. Transitional cases exhibit the lowest classification accuracy, consistent with the inherently gradual and parameter-sensitive nature of the intermediate regime.

The Logistic Regression standardised coefficients provide additional interpretable information. For the Fragmented class, the coefficient for Current (A) is negative (-0.809), indicating that increasing current reduces the probability of fragmented melting. Conversely, increasing Distance (mm) raises this probability ($+0.518$), consistent with the dispersed arc behaviour observed at larger stand-off distances. For the Continuous and Stable class, Distance carries the largest negative coefficient (-0.581), confirming that reduced stand-off is a necessary condition for stable melting. These coefficient signs align with the physical

interpretation presented throughout Section 4, lending statistical support to the qualitative parameter hierarchy described therein.

Permutation importance analysis (Figure 8f) consistently identifies applied current as the most influential predictor in both models (DT: 0.464 ± 0.117 ; LR: 0.274 ± 0.113), followed by electrode-to-powder-bed distance (DT: 0.210 ± 0.091 ; LR: 0.154 ± 0.091). Scanning speed shows near-zero importance under both models across the full parameter space investigated, which reflects the fact that the speed range in the present dataset is insufficient to dominate over the current and distance effects at a global level. The process maps derived from the Logistic Regression boundary (Figure 8d,e) project the predicted stability regions onto two-dimensional parameter planes, providing a continuous visual representation of the estimated classification boundaries consistent with the experimental observations.

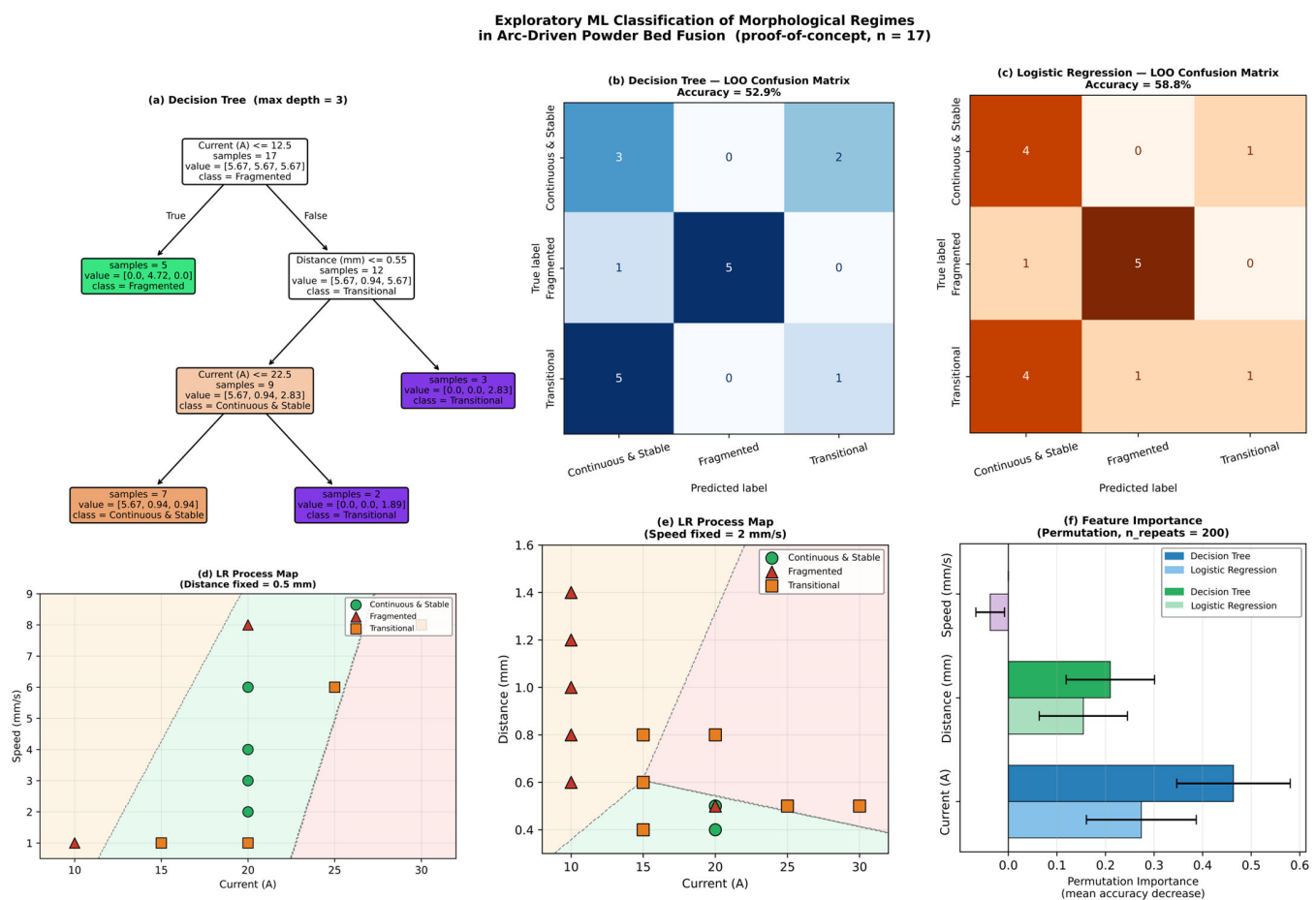


Figure 8. Exploratory machine learning classification of morphological regimes in arc-driven powder bed fusion ($n = 17$). (a) Decision Tree structure (max depth = 3); (b) Decision Tree leave-one-out confusion matrix (LOO-CV accuracy = 52.9%); (c) Logistic Regression leave-one-out confusion matrix (LOO-CV accuracy = 58.8%); (d) Logistic Regression process stability map in the Current–Speed plane at a fixed electrode distance of 0.5 mm; (e) Logistic Regression process stability map in the Current–Distance plane at a fixed scanning speed of 2 mm s^{-1} ; (f) permutation feature importance for both classifiers ($n = 200$ repeats). Background colours indicate predicted regime regions: green = Continuous and Stable; orange = Transitional; red = Fragmented. Experimental observations are overlaid as scatter markers.

4. Discussion

4.1. Influence of Applied Current and Scanning Speed on Melting Behaviour

Applied current is one of the dominant parameters governing heat input and melting behaviour in arc-driven powder bed processing. As the applied current increases, a greater amount of thermal energy is delivered to the powder bed, leading to an expansion of the melted region, as observed in the surface morphologies presented in Figure 3. Similar relationships between current level and melt dimensions have been reported in arc-based additive manufacturing studies, where increased current was associated with larger melted regions and enhanced material interaction [36].

At elevated current levels, increased surface disturbances such as particle agglomeration and localised surface degradation were observed in the present experiments, particularly at applied currents of 20 A and above. Comparable surface instabilities at high heat input levels have been documented in previous arc-driven melting and deposition studies, where excessive energy input promoted irregular surface features and non-uniform solidification behaviour [37–39]. In contrast, at lower current levels and reduced electrode-to-powder-bed distances, the arc appeared more concentrated, resulting in a more localised thermal input and comparatively stable melting behaviour. Similar observations have been reported in micro-scale TIG-based processing studies, where controlled current levels contributed to improved melting stability and surface quality [40].

Scanning speed plays a critical role in determining the duration of interaction between the arc and the powder bed. As the scanning speed increases, the effective interaction time decreases, limiting the amount of thermal energy delivered to a given location. This effect is clearly reflected in the experimental results presented in Figure 7 and Table 5, where both line width and melt width decrease progressively with increasing scanning speed. At scanning speeds exceeding approximately 6–8 mm/s, the continuity of the melted region deteriorated, and partially solidified segments were observed along the melting path. Similar behaviour has been reported in arc-based additive manufacturing studies, where high travel speeds were shown to disrupt continuous melting and promote premature solidification [41–43].

From an energy perspective, the combined influence of applied current and scanning speed can be interpreted using a relative line energy approach, in which the effective energy input per unit length scales with the ratio of applied current to scanning speed. Since the TIG power source operates in constant-current mode and the arc voltage varies dynamically with arc length and surface conditions, energy-related effects are discussed here on a comparative rather than absolute basis. The use of relative energy indicators or current-to-speed ratios is well established in arc-based additive manufacturing literature, where dynamic voltage behaviour makes fixed power calculations impractical [44].

Beyond geometric melt pool enlargement, the quantitative image-based analysis presented in Tables 3 and 5 demonstrates that energy input governs the topological continuity and statistical stability of solidification. Increasing current and decreasing stand-off distance systematically reduce the number of disconnected components (N), indicating progressive topological simplification of the melt structure. Binary phase continuity measurements are widely accepted as reliable indicators of structural connectivity in material systems [28]. The reduction in N from highly fragmented regimes to a single connected component confirms that increasing effective energy density drives the system from discontinuous solidification toward continuous melt track formation.

For conditions satisfying strict continuity ($N = 1$), width-based statistical analysis further reveals the stabilisation mechanism. The decrease in width dispersion (CV (width)) below the 0.15 threshold indicates geometric homogenization of the melt line. Variance-based descriptors have long been emphasised in quantitative microstructural characterisation

as indicators of process stability and deterministic growth behaviour [45]. The observed reduction in width dispersion with increasing effective energy density confirms that stabilisation is not merely geometric enlargement but a statistically measurable transition toward uniform melting.

The influence of scanning speed extends this interpretation. At low scanning speeds, sufficient interaction time allows for conductive heat transfer and stable coalescence, resulting in both topological continuity and low width dispersion. As scanning speed increases, interaction time decreases, promoting intermittent solidification and fragmentation. Under high-speed conditions, even increased current cannot fully compensate if interaction time remains insufficient, demonstrating that melting behaviour is governed by the balance between applied current and scanning speed rather than by either parameter independently.

4.2. Role of Electrode-to-Powder-Bed Distance on Melting Behaviour

The electrode-to-powder-bed distance was identified as a critical geometric parameter governing melting behaviour in the arc-driven powder bed system. As demonstrated by the experimental results presented in Section 3.2, variations in this distance significantly influenced both surface melting characteristics and melt pool cross-sectional features (Figures 4 and 5). In contrast to applied current and scanning speed, which primarily affect the magnitude of energy input, the electrode distance directly controls the spatial distribution and effective interaction of the arc with the powder bed.

At relatively large electrode-to-powder-bed distances, surface observations revealed discontinuous melting behaviour characterised by discrete agglomerated regions rather than continuous linear features. The corresponding cross-sectional observations indicated limited penetration and fragmented molten zones, suggesting insufficient coupling between the arc and the powder bed under these conditions. Similar distance-dependent limitations have been reported in arc-based processing studies, where excessive stand-off distances were shown to reduce arc concentration and hinder effective energy transfer to the substrate or feedstock [38–40].

Reducing the electrode-to-powder-bed distance led to a marked improvement in melting continuity. As shown in Figure 5, linear melted regions first appeared at intermediate distances, accompanied by more clearly defined molten zones in the corresponding cross-sections (Figure 6). This transition indicates that a sufficiently small stand-off distance is required to stabilise the arc–powder interaction and enable sustained linear melting. Comparable observations have been reported in micro-scale arc-based deposition and surface modification studies, where controlled electrode positioning was found to enhance process stability and melting consistency [41,42].

At the smallest investigated electrode-to-powder-bed distances, continuous linear melting was achieved over a wider range of applied current levels. However, the experimental results also showed that excessively small distances introduced surface irregularities and local instabilities along the melted track, as evidenced by intermittent ruptures and multiple peripheral melting features (Figure 5g,h). Similar trade-offs between arc concentration and process stability have been discussed in previous arc-driven manufacturing studies, highlighting the sensitivity of the process to electrode positioning [46].

From a geometric and energetic standpoint, these observations indicate that electrode-to-powder-bed distance governs the effective footprint of the arc on the powder bed and thereby modulates the local energy distribution. While applied current and scanning speed determine the overall energy input, the electrode distance defines how efficiently this energy is coupled into the powder bed. Consequently, the electrode-to-powder-bed distance emerges as a key parameter for achieving stable single-track melting, particularly during the early stages of arc-driven powder bed processing.

In addition to the geometrical melt pool observations, the quantitative image-based results presented in Table 3 provide further insight into the role of electrode distance in governing morphological stability. As the electrode-to-powder-bed distance decreases, the number of disconnected components (N) decreases systematically, indicating progressive topological simplification of the solidified structure. The reduction in N from highly fragmented regimes toward a single connected component confirms that decreasing stand-off distance enhances arc concentration and promotes deterministic coalescence.

For conditions approaching continuity, width-based statistical analysis further clarifies the stabilisation mechanism. When N reaches unity, the evaluation of width dispersion (CV (width)) reveals whether geometric uniformity is simultaneously achieved. The stabilisation observed at reduced stand-off distances is therefore not only visual but statistically measurable. Variance-based morphological descriptors are widely used to assess structural uniformity in materials characterisation [47], and reduced width dispersion under controlled geometric variation reflects improved process stability.

The transition from fragmented to continuous melting observed in Figure 5 and quantified in Table 3 suggests the presence of a threshold-type response in arc–powder coupling efficiency. Once a critical stand-off distance is reached, the arc becomes sufficiently constricted to sustain uninterrupted linear melting. Similar energy-coupling threshold phenomena have been discussed in additive manufacturing systems where geometric parameters strongly influence local energy concentration.

4.3. Combined Effects of Scanning Speed and Electrode Distance: Process Limitations

The results presented in Section 3.3 demonstrate that scanning speed represents a critical limiting parameter for maintaining continuous melting in the arc-driven powder bed system, even when favourable electrode-to-powder-bed distances are employed. At the selected electrode distance identified in Section 3.2, increasing the scanning speed resulted in a progressive reduction in both line width and melt width, ultimately leading to discontinuous melting behaviour (Figure 7, Table 5). These observations indicate that scanning speed imposes an upper limit on the stability of single-track melting under the investigated conditions.

The combined influence of scanning speed and electrode-to-powder-bed distance highlights the interdependence of geometric and temporal factors in arc-driven powder bed processing. While reducing the electrode distance enhances arc coupling and promotes more efficient local energy delivery to the powder bed (Figures 4 and 5), this improvement cannot fully offset the reduced interaction time associated with higher scanning speeds. Similar limitations have been reported in arc-based additive manufacturing studies, where increased travel speeds were shown to restrict continuous material melting despite favourable energy input conditions [41–43].

From a relative energy standpoint, the observed behaviour can be interpreted by considering the balance between effective energy input per unit length and interaction time. Although higher applied current levels increase the overall energy available, the reduction in interaction time at elevated scanning speeds limits the extent to which this energy can be effectively transferred into the powder bed. This interpretation is consistent with previous arc-based studies employing relative energy or current-to-speed ratio concepts to describe process boundaries under constant-current operating conditions [46].

The quantitative image-based results presented in Table 5 reinforce this limitation. At low scanning speeds, several conditions satisfy the strict continuity requirement ($N = 1$), and width dispersion remains below the defined stability threshold (CV (width)) < 0.15). As scanning speed increases, the number of disconnected components increases, and width variation exceeds the uniformity criterion. Even under reduced electrode distances that

promote enhanced arc coupling, higher travel speeds lead to topological fragmentation and increased geometric dispersion.

This behaviour confirms that geometric concentration of the arc (via reduced electrode distance) and temporal energy availability (via scanning speed) act as coupled but non-equivalent control parameters. Electrode distance primarily governs spatial energy distribution and arc–powder coupling efficiency, whereas scanning speed governs temporal energy exposure. Continuous melting is achieved only when sufficient arc concentration coincides with adequate interaction time.

The sharp transition from continuous to discontinuous melting observed at elevated scanning speeds suggests the presence of a process boundary defined by insufficient interaction time rather than insufficient arc concentration. Similar threshold-type limitations have been discussed in additive manufacturing systems where spatial and temporal energy factors jointly determine stability Windows [47].

Therefore, the practical processing window for arc-driven powder bed systems can be understood as a coupled stability domain bounded by geometric (electrode distance) and temporal (scanning speed) constraints. Stable melting is achieved only when both arc concentration and interaction time satisfy the continuity and width-uniformity criteria defined in Section 2. Outside this domain, discontinuous melting and topological fragmentation become unavoidable, regardless of isolated parameter optimisation.

From a control perspective, the monotonic variation in connected component number (N) and width dispersion ($CV(\text{width})$) across parameter combinations indicates a deterministic energy-driven regime transition rather than random fluctuation. The observed behaviour reflects a stability–instability boundary governed by the balance between spatial energy concentration and temporal exposure in arc-driven powder bed processing.

4.4. Exploratory Machine Learning Perspective

The exploratory ML analysis presented in Section 3.5 is positioned as a data-driven complement to the thermophysical interpretation, consistent with recent ML applications in arc-based and laser-based additive manufacturing [19,20]. In those studies, ML models were shown to provide quantitative insights into parameter dominance that are difficult to extract from experimental observations alone. The present work applies a similar philosophy to arc-driven powder bed fusion, a process configuration that has not previously been examined through a machine learning lens. The LOO-CV accuracies of 52.9% (DT) and 58.8% (LR) exceed the three-class random baseline of 33.3%. For $n = 17$, 52.9% accuracy corresponds to 9 correct predictions versus approximately 6 expected by chance. The permutation importance ranking (Current > Distance > Speed) is fully consistent across both classifiers, providing model-agnostic, quantitative corroboration of the thermophysical parameter hierarchy. To extend the ML analysis beyond classification, two linear regression models were fitted as first-order surrogate models for melt pool geometry. Melt pool depth ($n = 9$, cross-sectional micrographs) yielded $R^2(\text{LOO}) = 0.790$ and $\text{MAE} = 0.057$ mm (Table 9). Surface track width ($n = 9$, binary top-view images with pixel-to-mm calibration based on 10×30 mm substrate) yielded $R^2(\text{LOO}) = 0.696$ and $\text{MAE} = 0.187$ mm. Scanning speed is the strongest predictor of width variation; applied current dominates depth—consistent with the thermophysical interpretation. These models are valid only within the investigated parameter ranges and assume linear additive effects; their primary value lies in process map generation and rapid parameter screening. Results are summarised in Tables 9 and 10.

These first-order surrogate models are valid only within the investigated parameter ranges, assume linear additive effects, and do not capture interaction terms or threshold-type transitions. Their primary value lies in process map generation and rapid parameter screening rather than predictive generalisation beyond the experimental window.

Table 9. Linear regression model performance for melt pool geometry estimation (LOO-CV, $n = 9$). MAE: Mean Absolute Error.

Response Variable	n	R^2 (Full)	R^2 (LOO-CV)	MAE (LOO-CV)	Intercept
Melt Pool Depth (mm)	9	0.928	0.790	0.057 mm	0.291
Surface Track Width (mm)	9	0.930	0.696	0.187 mm	1.136

Table 10. Standardised regression coefficients (β) for melt pool depth and width models. Predictors standardised to zero mean and unit variance. (Positive distance coefficient reflects lateral arc-footprint spreading at larger stand-off distances.) (✓: coefficient sign consistent with physical expectation. *: positive coefficient due to lateral arc-footprint spreading.)

Parameter	β (Depth)	β (Width)	Sign Check	Physical Interpretation
Current (A)	+0.133	+0.397	✓ Both +	Higher current → more thermal energy → larger melt pool
Electrode Distance (mm)	+0.104	+0.185	*	Larger distance → dispersed arc footprint → wider but shallower pool *
Scanning Speed (mm/s)	−0.177	−0.573	✓ Both −	Higher speed → reduced interaction time → smaller melt pool

Taken together, the classification and regression results position the ML component as a structured, data-driven contribution to process understanding in arc-driven powder bed fusion: classification models provide a statistically supported parameter hierarchy, while regression models offer first-order predictive capability for melt pool geometry—enabling preliminary process map construction and rapid parameter screening. Future work involving larger experimental datasets will enable more robust predictive models and uncertainty quantification, moving the framework progressively from exploratory analysis toward engineering-grade process optimisation tools.

The permutation importance ranking—Current > Distance > Speed—is fully consistent with the thermophysical interpretations developed in Sections 4.1–4.3 and provides quantitative, model-agnostic corroboration of the qualitative parameter hierarchy reported there. Applied current governs the total thermal energy available for melting, and its dominant importance score (0.464 for DT, 0.274 for LR) reflects the fact that no stable melting is achievable below a current threshold, regardless of the other parameters. The secondary importance of electrode distance confirms that spatial energy concentration is a necessary geometric condition for regime stabilisation once the thermal threshold is exceeded. The near-zero importance of scanning speed at the global dataset level reflects the narrower speed range investigated relative to the current and distance ranges, rather than a physical insignificance of this parameter—a distinction that is also consistent with the focused speed sensitivity analysis discussed in Section 4.3.

The Logistic Regression coefficient signs further corroborate the physical interpretation. The negative coefficient of Current for the Fragmented class (−0.809) quantitatively confirms that increasing current systematically shifts the system away from fragmented melting. The negative coefficient of Distance for the Continuous and Stable class (−0.581) reflects the experimentally observed requirement for reduced stand-off to achieve arc constriction sufficient for continuous melt formation. Taken together, these statistical indicators are coherent with the mechanistic description developed throughout this study, and their alignment with the experimental findings supports the internal consistency of the overall analytical framework, notwithstanding the exploratory nature of the ML component.

4.5. Scope and Limitations of the Present Study

The present work focuses on the investigation of early-stage melting behaviour in an arc-driven powder bed system through controlled single-track experiments. The effects of applied current, scanning speed, and electrode-to-powder-bed distance were systematically

examined to identify the key sensitivities governing the formation and stabilisation of linear melted regions. By limiting the analysis to single-track melting, the study provides a simplified and controlled framework for understanding the fundamental behaviour of arc–powder interaction during the initial stages of process development.

Energy-related effects were interpreted using relative indicators derived from applied current and scanning speed rather than absolute power or volumetric energy density values. This approach reflects the constant-current operating mode of the TIG power source and the dynamic variation in arc voltage during interaction with a powder bed surface. Accordingly, the energy perspective adopted here supports comparative interpretation of parameter trends rather than absolute quantification of process efficiency or optimisation limits.

The investigation does not address multi-track interactions, layer-by-layer deposition, or cumulative thermal effects associated with extended processing. In practical additive manufacturing environments, repeated layer deposition may introduce thermal accumulation, residual stress development, and melt pool overlap effects that modify local stability conditions. Similarly, operational factors such as powder reuse, shielding gas flow variation, and long-term arc stability were not included in the present parameter window. These aspects are expected to influence process robustness and represent important directions for future work.

In addition, the quantitative image-based framework employed in this study is restricted to two-dimensional representations of surface morphology and cross-sectional geometry. While the connected component number (N) and width-based dispersion metric (CV (width)) provide reproducible and deterministic descriptors of morphological evolution, they do not capture three-dimensional melt pool dynamics, transient fluid flow behaviour, or thermocapillary-driven instabilities. Extension of the present regime interpretation to volumetric process modelling would require integration with in situ monitoring techniques and thermo-fluid simulation approaches.

Furthermore, the proposed two-stage continuity–uniformity framework is formulated for comparative regime identification within the investigated parameter window. Although monotonic variations in N and (CV (width)) demonstrate strong correlation with energy-related parameters, predictive generalisation beyond the studied range requires broader parameter mapping. The establishment of universal threshold criteria for arc-driven powder bed systems would necessitate expanded experimental datasets and potentially coupling with physics-based thermal models.

Despite these limitations, the present study establishes a structured methodology for linking geometric process parameters with statistically quantifiable morphological transitions. The combined experimental and image-based approach provides a reproducible basis for defining stability boundaries in arc-driven powder bed processing and offers a foundation for future investigations involving multi-track deposition, multilayer fabrication, and model-based optimisation strategies.

Beyond the experimental extensions outlined above, the interpretability framework developed in this study—comprising permutation-based feature importance rankings and standardised logistic regression coefficients—provides a structured basis for integration with regression-based process optimisation approaches. Feature importance scores identify which parameters warrant priority attention in optimisation campaigns, while the coefficient signs and magnitudes offer first-order guidance on the direction and relative strength of parameter effects. Future work should therefore explore how this explainability framework can be coupled with response surface methods or Gaussian process regression models to construct quantitative process maps with well-characterised stability boundaries, progressively moving the analytical framework from exploratory classification toward engineering-grade process optimisation.

5. Conclusions

A systematic experimental investigation was conducted to characterise the single-track melting behaviour of 316L stainless steel in an arc-driven powder bed system under controlled variation in applied current, scanning speed, and electrode-to-powder-bed distance. The results demonstrate that these parameters act in a coupled manner to govern melting continuity, melt pool geometry, and morphological stability. Applied current emerged as the primary determinant of thermal energy availability and regime transition. Low current levels produced fragmented, balling-dominant morphologies, while increasing current promoted progressive coalescence toward continuous melt track formation. Electrode-to-powder-bed distance governed arc constriction and spatial energy concentration: reducing stand-off distance enhanced arc–powder coupling efficiency and promoted deterministic coalescence, whereas excessive distances resulted in dispersed energy delivery and fragmented solidification. The only condition satisfying both stages of the deterministic framework—full topological continuity ($N = 1$) and geometric stability ($CV(\text{width}) = 0.128 < 0.15$)—was achieved at 20 A with a stand-off distance of 0.4 mm. Scanning speed governed temporal energy exposure: stable continuous tracks were consistently obtained within the 2–4 mm/s range at 20 A, while increasing speed beyond this range led to progressive geometric instability and ultimately topological fragmentation at 8 mm/s. The exploratory machine learning analysis corroborated the thermophysical parameter hierarchy in a quantitative, model-agnostic manner. LOO-CV accuracies of 52.9% (Decision Tree) and 58.8% (Logistic Regression) exceed the three-class random baseline of 33.3%. Permutation importance consistently ranked applied current as the dominant predictor (DT: 0.464; LR: 0.274), followed by electrode distance and scanning speed. Logistic Regression standardised coefficients confirmed that increasing current reduces fragmentation probability and that reduced stand-off distance is a necessary condition for stable arc constriction. Supplementary linear regression models yielded $R^2(\text{LOO}) = 0.790$ for melt pool depth and $R^2(\text{LOO}) = 0.696$ for surface track width, confirming that current dominates depth and scanning speed governs width variation. The combined framework establishes a reproducible methodology for morphological regime identification in arc-driven powder bed processing. Future work should expand the experimental dataset to enable robust model validation and uncertainty quantification, and should investigate the integration of the present explainability framework with regression-based and physics-informed approaches for quantitative process map construction and engineering-grade process optimisation.

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Abbreviations

The following abbreviations are used in this manuscript:

PBF	Powder Bed Fusion
WAAM	Wire Arc Additive Manufacturing
TIG	Tungsten Inert Gas
AM	Additive Manufacturing
N	Number of Connected Components
CV	Coefficient of Variation
CV (width)	Coefficient of Variation of Track Width
HAZ	Heat Affected Zone
PBF-Arc	Arc-Driven Powder Bed Fusion
mm	Millimeter
A	Ampere
ISO	International Organization for Standardization
ND	Neutral Density

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